

Part 2: Fourier Analysis

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BASIC CONCEPT OF FOURIER ANALYSIS

Most of the phenomena studied in the domain of Engineering and Science are periodic in nature. For instance, current and voltage in an alternating current circuit. These periodic functions could be analyzed into their constituent components (fundamentals and harmonics) by a process called **Fourier analysis**.

FOURIER SERIES

A Fourier series is an expansion of a periodic function $f(x)$ in terms of an infinite sum of sines and cosines. Fourier Series makes use of the orthogonality relationships of the sine and cosine functions.

The formula of Fourier series is given as:

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Again, if $f(x)$ is periodic from $(-L)$ to $(+L)$.

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

Now, from the formula of Fourier series, it is clear that, in order to find the series, we need to first know the values of a_0 , a_n and b_n . So, let's find their values.

For a_n :

From Fourier Series,

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Taking $\cos nx$ on both sides,

$$\cos nx \ f(x) = \cos nx \ a_0 + \cos nx \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Integrating both sides,

$$\int_{-\pi}^{\pi} \cos nx \ f(x) \ dx = \int_{-\pi}^{\pi} a_0 \cos nx \ dx + \int_{-\pi}^{\pi} \cos nx \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \ dx$$

$$\text{or, } \int_{-\pi}^{\pi} \cos nx \ f(x) \ dx = 0 + a_n \pi$$

$$\therefore a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \ dx$$

For b_n :

From Fourier Series,

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Taking $\sin nx$ on both sides,

$$\sin nx \ f(x) = \sin nx \ a_0 + \sin nx \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Integrating both sides,

$$\int_{-\pi}^{\pi} \sin nx \ f(x) \ dx = \int_{-\pi}^{\pi} a_0 \sin nx \ dx + \int_{-\pi}^{\pi} \sin nx \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \ dx$$

$$\text{or, } \int_{-\pi}^{\pi} \sin nx \ f(x) \ dx = 0 + b_n \pi$$

$$\therefore \ b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \ dx$$

For a_0 :

From Fourier Series,

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

Taking $\cos 0x$ on both sides,

$$\cos 0x \ f(x) = \cos 0x \ a_0 + \cos 0x \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

Integrating both sides,

$$\int_{-\pi}^{\pi} \cos 0x \ f(x) \ dx = \int_{-\pi}^{\pi} a_0 \cos 0x \ dx + \int_{-\pi}^{\pi} \cos 0x \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right) \ dx$$

$$\text{or, } \int_{-\pi}^{\pi} f(x) \ dx = a_0 [x]_{-\pi}^{\pi} + 0$$

$$\therefore \ a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \ dx$$

So, in summary, the necessary formulae for any type of Fourier series dealing with REAL numbers are:

Fourier Series:

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Different parameters to find Fourier Series:

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

Example 22: If $f(t) = \begin{cases} -1 & ; -\pi < t < 0 \\ +1 & ; 0 < t < \pi \end{cases}$ & $f(t)$ is periodic over $-\pi$ to π , find the Fourier series of $f(t)$.

Soln.: From Euler's Formula,

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) dt$$

$$= \frac{1}{2\pi} \left[\int_{-\pi}^0 (-1) dt + \int_0^{\pi} (1) dt \right]$$

$$= \frac{1}{2\pi} [[t]_{-\pi}^0 + [t]_0^{\pi}]$$

$$= \frac{1}{2\pi} (-\pi + \pi)$$

$$= 0$$

Again, from Euler's Formula,

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos nt dt$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 (-1) \cos nt dt + \int_0^{\pi} (1) \cos nt dt \right]$$

$$= \frac{1}{\pi} \left[-\left[\frac{\sin nt}{n}\right]_{-\pi}^0 + \left[\frac{\sin nt}{n}\right]_0^{\pi} \right]$$

$$= \frac{1}{\pi} \times 0$$

$$= 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt \, dt$$

$$\begin{aligned} &= \frac{1}{\pi} \left[\int_{-\pi}^0 (-1) \sin nt \, dt + \int_0^{\pi} (1) \sin nt \, dt \right] \\ &= \frac{1}{\pi} \left[\left[\frac{\cos nt}{n} \right]_{-\pi}^0 + \left[\frac{\cos nt}{n} \right]_0^{\pi} \right] \\ &= \frac{1}{\pi} \left[\frac{1}{n} - \frac{\cos n\pi}{n} - \frac{\cos n\pi}{n} + \frac{1}{n} \right] \\ &= \frac{2}{n\pi} (1 - \cos n\pi) \\ &= \frac{2}{n\pi} [1 - (-1)^n] \end{aligned}$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt)$$

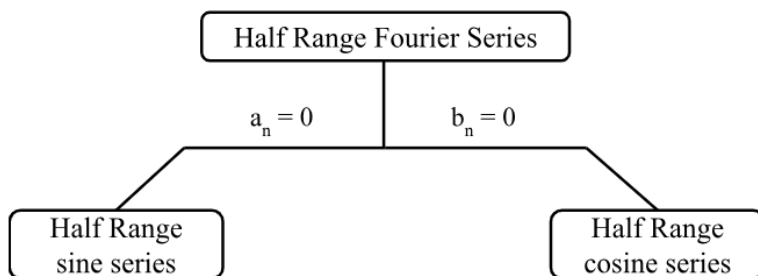
$$= 0 + \sum_{n=1}^{\infty} \left(0 + \frac{2}{n\pi} [1 - (-1)^n] \sin nt \right)$$

$$= \sum_{n=1}^{\infty} \frac{2}{n\pi} [1 - (-1)^n] \sin nt$$

$$= \frac{2}{\pi} \cdot 2 \sin t + 0 + \frac{2}{3\pi} \cdot 2 \sin 3t + 0 + \frac{2}{5\pi} \cdot 2 \sin 5t + \dots \dots \dots$$

$$= \frac{\pi}{4} \left(\sin t + \frac{\sin 3t}{3} + \frac{\sin 5t}{5} + \dots \dots \dots \right) \quad (\text{Ans.})$$

We can understand it clearly seeing the chart below:



Example 23: If $f(t) = t - t^2$; & $f(t)$ is periodic over $-\pi$ to π , find the Fourier series of $f(t)$.

Soln.: From Euler's Formula,

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) dt \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (t - t^2) dt \\ &= \frac{1}{2\pi} \left[\left[\frac{t^2}{2} \right]_{-\pi}^{\pi} - \left[\frac{t^3}{3} \right]_{-\pi}^{\pi} \right] \\ &= \frac{1}{2\pi} \left[\frac{\pi^2}{2} - \frac{\pi^2}{2} - \frac{\pi^3}{3} - \frac{\pi^3}{3} \right] \\ &= -\frac{\pi^2}{3} \end{aligned}$$

Again, from Euler's Formula,

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos nt dt \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} (t - t^2) \cos nt dt \dots \dots \dots (i) \end{aligned}$$

Now, [Performing the integration part first]

$$\begin{aligned} &\int (t - t^2) \cos nt dt \\ &= (t - t^2) \int \cos nt dt - \int \left[(1 - 2t) \int \cos nt dt \right] dt \\ &= (t - t^2) \frac{\sin nt}{n} - \int \left[(1 - 2t) \frac{\sin nt}{n} \right] dt \\ &= (t - t^2) \frac{\sin nt}{n} - \frac{1}{n} \left[(1 - 2t) \int \sin nt dt - \int \left\{ (-2) \int \sin nt dt \right\} dt \right] \\ &= (t - t^2) \frac{\sin nt}{n} - \frac{1}{n} \left[(1 - 2t) \left(-\frac{\cos nt}{n} \right) - \int \left\{ (-2) \left(-\frac{\cos nt}{n} \right) \right\} dt \right] \\ &= (t - t^2) \frac{\sin nt}{n} - \frac{1}{n} \left[-\frac{\cos nt}{n} + \frac{2t \cos nt}{n} - \frac{2 \sin nt}{n^2} \right] \\ &= \frac{t \sin nt}{n} - \frac{t^2 \sin nt}{n} + \frac{\cos nt}{n^2} - \frac{2t \cos nt}{n^2} + \frac{2 \sin nt}{n^3} \end{aligned}$$

From equation (i),

$$\begin{aligned}
 a_n &= \frac{1}{\pi} \left[\frac{t \sin nt}{n} - \frac{t^2 \sin nt}{n} + \frac{\cos nt}{n^2} - \frac{2t \cos nt}{n^2} + \frac{2 \sin nt}{n^3} \right]_{-\pi}^{\pi} \\
 &= \frac{1}{\pi} \left[\frac{(-1)^n}{n^2} - \frac{2\pi(-1)^n}{n^2} - \frac{(-1)^n}{n^2} + \frac{2(-\pi)(-1)^n}{n^2} \right] \\
 &= \frac{1}{\pi} \left[-\frac{4\pi(-1)^n}{n^2} \right] \\
 &= -\frac{4(-1)^n}{n^2}
 \end{aligned}$$

Lastly, from Euler's Formula,

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt \, dt \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} (t - t^2) \sin nt \, dt \dots \dots \dots (ii)
 \end{aligned}$$

Now, [Performing the integration part first]

$$\begin{aligned}
 &\int (t - t^2) \sin nt \, dt \\
 &= (t - t^2) \int \sin nt \, dt - \int \left[(1 - 2t) \int \sin nt \, dt \right] dt \\
 &= (t - t^2) \left(-\frac{\cos nt}{n} \right) - \int \left[(1 - 2t) \left(-\frac{\cos nt}{n} \right) \right] dt \\
 &= (t - t^2) \left(-\frac{\cos nt}{n} \right) + \frac{1}{n} \left[(1 - 2t) \int \cos nt \, dt - \int \left\{ (-2) \int \cos nt \, dt \right\} dt \right] \\
 &= (t - t^2) \left(-\frac{\cos nt}{n} \right) + \frac{1}{n} \left[(1 - 2t) \left(\frac{\sin nt}{n} \right) - \int \left\{ (-2) \left(\frac{\sin nt}{n} \right) \right\} dt \right] \\
 &= (t - t^2) \left(-\frac{\cos nt}{n} \right) + \frac{1}{n} \left[\frac{\sin nt}{n} - \frac{2t \sin nt}{n} + \frac{2 \cos nt}{n^2} \right] \\
 &= -\frac{t \cos nt}{n} + \frac{t^2 \cos nt}{n} + \frac{\sin nt}{n^2} - \frac{2t \sin nt}{n^2} + \frac{2 \cos nt}{n^3}
 \end{aligned}$$

From equation (ii),

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \left[-\frac{t \cos nt}{n} + \frac{t^2 \cos nt}{n} + \frac{\sin nt}{n^2} - \frac{2t \sin nt}{n^2} + \frac{2 \cos nt}{n^3} \right]_{-\pi}^{\pi} \\
 &= \frac{1}{\pi} \left[\frac{(-\pi)(-1)^n}{n} + \frac{\pi^2(-1)^n}{n} + \frac{2(-1)^n}{n^3} - \frac{\pi(-1)^n}{n} - \frac{\pi^2(-1)^n}{n} - \frac{2(-1)^n}{n^3} \right] \\
 &= \frac{1}{\pi} \left[-\frac{2\pi(-1)^n}{n} \right] \\
 &= -\frac{2(-1)^n}{n}
 \end{aligned}$$

We know, the Fourier series is:

$$\begin{aligned}
 f(t) &= a_0 + \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt) \\
 &= -\frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left[-\frac{4(-1)^n}{n^2} \cos nt - \frac{2(-1)^n}{n} \sin nt \right]
 \end{aligned}$$

(Ans.)

Practice Problems

1. If $f(x) = \begin{cases} 0 & ; -\pi \leq x < 0 \\ x & ; 0 \leq x < \pi \end{cases}$, find the Fourier series of $f(x)$ and hence evaluate $\sum_{n=1}^{\infty} \frac{1}{n^2}$.
2. If $f(x) = \begin{cases} -1 & ; -\pi \leq x < 0 \\ +1 & ; 0 \leq x \leq \pi \end{cases}$; find the Fourier series of $f(x)$ and hence deduce $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \dots$
3. Expand $f(x) = x, 0 < x < 2$ in a half range sine series.
4. If $0 < x < \pi$, show that $x(\pi - x) = \frac{\pi^2}{6} - \sum_{n=1}^{\infty} \left(\frac{\cos 2nx}{n^2} \right)$
5. Expand $f(x) = e^{-x}, 0 < x < 2\pi$ in a Fourier series.
6. Expand $f(x) = \begin{cases} 2 - x, & 0 < x < 4 \\ x - 6, & 4 < x < 8 \end{cases}$ in a Fourier series of period 8.
7. If $f(x) = x^2$ & $f(x)$ is periodic over 2π , find the Fourier series of $f(x)$ and hence evaluate $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$.
8. If $f(x) = \begin{cases} 0 & ; -\pi \leq x < 0 \\ \sin x & ; 0 \leq x \leq \pi \end{cases}$ & $f(x)$ is periodic over 2π , find the Fourier series of $f(x)$.
9. If $f(x) = \begin{cases} -x & ; -\pi \leq x < 0 \\ +x & ; 0 \leq x \leq \pi \end{cases}$; find the Fourier series of $f(x)$.

10. Sketch 3 cycles of the function represented by:

$$f(t) = \begin{cases} 0 & ; -1 \leq t \leq -\frac{1}{2} \\ \cos 3\pi t & ; -\frac{1}{2} \leq t \leq \frac{1}{2} \\ 0 & ; \frac{1}{2} \leq t \leq 1 \end{cases} \text{ and } f(t) = f(t + 2)$$

Find the Fourier Series of the above function (up to 5 terms at least).

11. Find the constant term a_0 in the Fourier series corresponding to $f(x) = x - x^3$ in $(-\pi, \pi)$. And what kind of function is it? Draw a graph.

12. Using $\int_{-\pi}^{+\pi} \sin nx \, dx = 0$ and $\int_{-\pi}^{+\pi} \cos nx \, dx = 0$, where $m, n \in \mathbb{Z}$. Prove the following identities:

$$\text{a) } \int_{-\pi}^{\pi} \cos nx \cos mx \, dx = \begin{cases} 2\pi & \text{if } n = m = 0 \\ \pi & \text{if } n = m \neq 0 \\ 0 & \text{if } n \neq m \end{cases}$$

$$\text{b) } \int_0^{\pi} \cos nx \cos mx \, dx = \begin{cases} \pi & \text{if } n = m = 0 \\ \frac{\pi}{2} & \text{if } n = m \neq 0 \\ 0 & \text{if } n \neq m \end{cases}$$

$$\text{c) } \int_{-\pi}^{\pi} \sin nx \sin mx \, dx = \begin{cases} \pi & \text{if } n = m \\ 0 & \text{if } n \neq m \end{cases}$$

$$\text{d) } \int_0^{\pi} \sin nx \sin mx \, dx = \begin{cases} \frac{\pi}{2} & \text{if } n = m \\ 0 & \text{if } n \neq m \end{cases}$$

$$\text{e) } \int_{-\pi}^{\pi} \sin nx \cos mx \, dx = 0$$

13. Draw sketches and determine the Fourier Series for the following functions:

$$\text{a) } s(x) = \frac{x}{\pi}, \text{ for } -\pi < x < \pi$$

$$\text{b) } s(x) = 3|\sin x|, \text{ for } 0 \leq x \leq 2\pi$$

$$\text{c) } s(x) = \begin{cases} 2 \sin x & \text{for } 0 \leq x < \pi \\ 0 & \text{for } \pi \leq x \leq 2\pi \end{cases}$$

$$\text{d) } s(x) = \begin{cases} 1 & \text{for } 0 \leq x \leq \pi \\ 0 & \text{for } \pi \leq x \leq 2\pi \end{cases}$$

$$\text{e) } s(x) = A - \frac{Ax}{P} \text{ for } 0 \leq x < P$$

COMPLEX FORM OF FOURIER SERIES

From MAT215 we know that,

$$e^{i\theta} = \cos \theta + i \sin \theta \dots \dots \dots (i)$$

$$e^{-i\theta} = \cos \theta - i \sin \theta \dots \dots \dots (ii)$$

Adding equation (i) and (ii),

$$e^{i\theta} + e^{-i\theta} = \cos \theta + i \sin \theta + \cos \theta - i \sin \theta$$

$$e^{i\theta} + e^{-i\theta} = 2 \cos \theta$$

$$\therefore \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

Subtracting equation (i) and (ii),

$$e^{i\theta} - e^{-i\theta} = \cos \theta + i \sin \theta - \cos \theta + i \sin \theta$$

$$e^{i\theta} - e^{-i\theta} = 2i \sin \theta$$

$$\therefore \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$= \frac{-i(e^{i\theta} - e^{-i\theta})}{2}$$

$$\therefore \sin \theta = \frac{ie^{-i\theta} - ie^{i\theta}}{2}$$

Now, from the Fourier Series,

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt)$$

$$= a_0 + \sum_{n=1}^{\infty} \left(a_n \frac{e^{int} + e^{-int}}{2} + b_n \frac{ie^{-int} - ie^{int}}{2} \right)$$

$$= a_0 + \sum_{n=1}^{\infty} \left(\frac{a_n}{2} - \frac{ib_n}{2} \right) e^{int} + \sum_{n=1}^{\infty} \left(\frac{a_n}{2} + \frac{ib_n}{2} \right) e^{-int}$$

$$= a_0 + \sum_{n=1}^{\infty} \left(\frac{a_n}{2} - \frac{ib_n}{2} \right) e^{int} + \sum_{n=-\infty}^{-1} \left(\frac{a_n}{2} + \frac{ib_n}{2} \right) e^{int}$$

$$= a_0 e^{int}|_{n=0} + \sum_{n=1}^{\infty} \left(\frac{a_n}{2} - \frac{ib_n}{2} \right) e^{int} + \sum_{n=-\infty}^{-1} \left(\frac{a_n}{2} + \frac{ib_n}{2} \right) e^{int}$$

$$= \sum_{n=-\infty}^{\infty} c_n e^{int}$$

Therefore, the complex form of Fourier Series is: $f(t) = \sum_{n=-\infty}^{\infty} c_n e^{int}$

Now, in order to determine the series, we need to find the value of the coefficient c_n .

We got from the complex form of Fourier Series,

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{int}$$

Taking e^{-int} on both sides of the equation,

$$e^{-int} \cdot f(t) = e^{-int} \cdot \sum_{n=-\infty}^{\infty} c_n e^{int}$$

Integrating both sides with respect to t,

$$\int_{-\pi}^{\pi} e^{-int} \cdot f(t) dt = \int_{-\pi}^{\pi} e^{-int} \cdot \sum_{n=-\infty}^{\infty} c_n e^{int}$$
$$\text{or, } \int_{-\pi}^{\pi} f(t) \cdot e^{-int} dt = c_n \cdot 2\pi$$

$$\therefore c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) \cdot e^{-int} dt$$

So, in summary, the necessary formulae for Complex Form of Fourier series are:

Fourier Series:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{int}$$

Different parameters to find Fourier Series:

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) \cdot e^{-int} dt$$

The complex form of Fourier series is algebraically simpler and more symmetric. This is why, in Engineering Mathematics, the complex form of Fourier Series is most commonly used.

We will understand it even more clearly after completing the Parseval's Theorem.

Example 24: If $f(t) = \begin{cases} -1 & ; -\pi < t < 0 \\ +1 & ; 0 < t < \pi \end{cases}$ & $f(t)$ is periodic over 2π , find the Fourier series of $f(t)$ using the complex form.

Soln.: We know, the Complex Form of Fourier Series is given as:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{int}$$

From Euler's Formula,

$$\begin{aligned} c_n &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) \cdot e^{-int} dt \\ &= \frac{1}{2\pi} \left[\int_{-\pi}^0 (-1) e^{-int} dt + \int_0^{\pi} 1 \cdot e^{-int} dt \right] \\ &= \frac{1}{2\pi} \left(- \left[\frac{e^{-int}}{-in} \right]_{-\pi}^0 + \left[\frac{e^{-int}}{-in} \right]_0^{\pi} \right) \\ &= \frac{1}{2\pi} \left\{ \left(\frac{1}{in} - \frac{e^{in\pi}}{in} \right) - \left(\frac{e^{-in\pi}}{in} - \frac{1}{in} \right) \right\} \\ &= \frac{1}{2in\pi} (1 - e^{in\pi} - e^{-in\pi} + 1) \\ &= \frac{1}{2in\pi} \{2 - (e^{in\pi} + e^{-in\pi})\} \\ &= \frac{1}{2in\pi} (2 - 2 \cos n\pi) \\ &= \frac{1}{in\pi} (1 - \cos n\pi) \\ &= -\frac{i}{n\pi} \{1 - (-1)^n\} \end{aligned}$$

Now, substituting the value of c_n in the Complex Form of Fourier Series,

$$\begin{aligned} f(t) &= \sum_{n=-\infty}^{\infty} c_n e^{int} \\ &= \sum_{n=-\infty}^{\infty} -\frac{i}{n\pi} \{1 - (-1)^n\} e^{int} \\ &= \sum_{n=-\infty}^{\infty} \frac{i}{n\pi} \{(-1)^n - 1\} e^{int} \end{aligned}$$

(Ans.)

Practice Problem

1. If $f(t) = t - t^2$; & $f(t)$ is periodic over 2π , find the Fourier series of $f(t)$ using the complex form.

PARSEVAL'S IDENTITY

We know, the Fourier series dealing with real numbers is given as:

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt)$$

The Parseval's theorem deals with the square of the Fourier series, i.e., $\{f(t)\}^2$. It is like the **Pythagoras Theorem** for Fourier series.

Before we know about the Parseval's identity, it is important to recall some of the basic integral concepts.

1. $\int_{-\pi}^{\pi} \sin nx \cdot \sin mx \, dx = \begin{cases} 0 & \text{when, } m \neq n \\ \pi & \text{when, } m = n \end{cases}$
2. $\int_{-\pi}^{\pi} \cos nx \cdot \cos mx \, dx = \begin{cases} 0 & \text{when, } m \neq n \\ \pi & \text{when, } m = n \end{cases}$
3. $\int_{-\pi}^{\pi} \sin nx \cdot \cos mx \, dx = 0$

The Parseval's identity states that,

$$\frac{1}{\pi} \int_{-\pi}^{\pi} \{f(t)\}^2 \, dt = a_0^2 + \sum_{n=1}^{\infty} a_n^2 + b_n^2$$

Let's prove the statement:

We know,

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt)$$

Squaring both sides,

$$\{f(t)\}^2 = \left\{ a_0 + \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt) \right\}^2$$

Integrating both sides with respect to t,

$$\begin{aligned} \int_{-\pi}^{\pi} \{f(t)\}^2 \, dt &= \int_{-\pi}^{\pi} \left\{ a_0 + \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt) \right\}^2 \, dt \\ &= \int_{-\pi}^{\pi} \left\{ a_0^2 + 2a_0 \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt) + \left(\sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt) \right)^2 \right\} \, dt \\ &= \int_{-\pi}^{\pi} a_0^2 \, dt + \int_{-\pi}^{\pi} 2a_0 \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt) \, dt + \int_{-\pi}^{\pi} \left(\sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt) \right)^2 \, dt \\ &= a_0^2 [t]_{-\pi}^{\pi} + 2a_0 \int_{-\pi}^{\pi} \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt) \, dt + \int_{-\pi}^{\pi} \left(\sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt) \right) \\ &\quad \cdot \left(\sum_{m=1}^{\infty} (a_m \cos mt + b_m \sin mt) \right) \, dt \end{aligned}$$

$$\begin{aligned}
&= 2\pi a_0^2 + 2a_0 \cdot 0 + \int_{-\pi}^{\pi} \left(\sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt) \right) \cdot \left(\sum_{m=1}^{\infty} (a_m \cos mt + b_m \sin mt) \right) dt \\
&= 2\pi a_0^2 + \int_{-\pi}^{\pi} \left\{ \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} (a_n \cos nt \cdot a_m \cos mt + a_n \cos nt \cdot b_m \sin mt + b_n \sin nt \cdot a_m \cos mt + b_n \sin nt \cdot b_m \sin mt) \right\} dt \\
&= 2\pi a_0^2 + \sum_{n=1}^{\infty} a_n^2 \pi + \sum_{n=1}^{\infty} b_n^2 \pi \\
&= 2\pi a_0^2 + \sum_{n=1}^{\infty} (a_n^2 \pi + b_n^2 \pi)
\end{aligned}$$

$$or, \int_{-\pi}^{\pi} \{f(t)\}^2 dt = 2\pi a_0^2 + \sum_{n=1}^{\infty} (a_n^2 \pi + b_n^2 \pi)$$

$$\therefore \frac{1}{\pi} \int_{-\pi}^{\pi} \{f(t)\}^2 dt = 2a_0^2 + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

FOURIER TRANSFORMATION

Fourier Transformation is a mathematical model which helps to transform the signals between two different domains, such as transforming signal from frequency domain to time domain or vice versa. The generalization of the complex Fourier series is known as the Fourier transformation. The term “Fourier transformation” can be used in the mathematical function, and it is also used in the representation of the frequency domain. The Fourier transformation helps to extend the Fourier series to the non-periodic functions, which helps us to view any functions in terms of the sum of simple sinusoids. Fourier transformation has many applications in Engineering and Physics, such as signal processing, RADAR, and so on.

Fourier Transformation can be of two types.

- i. Forward Fourier Transformation, $F.T. \{f(t)\}$
- ii. Inverse Fourier Transformation, $I.F.T. \{F(\omega)\}$

The formulas for both types of Fourier Transformations are given as:

$$\begin{aligned}
F.T. \{f(t)\} &= F(\omega) = \int_{-\infty}^{\infty} f(t) \cdot e^{i\omega t} dt \\
I.F.T. \{F(\omega)\} &= f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) \cdot e^{-i\omega t} d\omega
\end{aligned}$$

Example 25: If $f(t) = \begin{cases} 1 & \text{for } |t| < 1 \\ 0 & \text{for } |t| > 1 \end{cases}$; find the Fourier Transformation of $f(t)$, hence evaluate $\int_0^\infty \frac{\sin t}{t} dt$.

Soln.: We know,

$$\begin{aligned}
 F.T. \{f(t)\} = F(\omega) &= \int_{-\infty}^{\infty} f(t) \cdot e^{i\omega t} dt \\
 &= \int_{-\infty}^{-1} f(t)e^{i\omega t} dt + \int_{-1}^1 f(t)e^{i\omega t} dt + \int_1^{\infty} f(t)e^{i\omega t} dt \\
 &= 0 + \int_{-1}^1 f(t)e^{i\omega t} dt + 0 \\
 &= \int_{-1}^1 1 \cdot e^{i\omega t} dt \\
 &= \left[\frac{e^{i\omega t}}{i\omega} \right]_{-1}^1 \\
 &= \frac{e^{i\omega t}}{i\omega} - \frac{e^{-i\omega t}}{i\omega} \\
 &= \frac{1}{\omega} \left(\frac{e^{i\omega} - e^{-i\omega}}{i} \right) \\
 &= \frac{1}{\omega} \times 2 \sin \omega \\
 &= \frac{2 \sin \omega}{\omega}
 \end{aligned}$$

$$\therefore F(\omega) = \frac{2 \sin \omega}{\omega}$$

Now,

$$\begin{aligned}
 I.F.T. \{F(\omega)\} = f(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) \cdot e^{-i\omega t} d\omega \\
 \text{or, } f(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2 \sin \omega}{\omega} \cdot e^{-i\omega t} d\omega \\
 &= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin \omega}{\omega} \cdot e^{-i\omega t} d\omega
 \end{aligned}$$

Therefore, $f(0) = 1 = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin \omega}{\omega} d\omega$ [As given condition says, At $t = 0$, $f(t) = 1$]

$$\text{or, } \int_{-\infty}^{\infty} \frac{\sin \omega}{\omega} d\omega = \pi$$

$$\text{or, } 2 \int_0^{\infty} \frac{\sin \omega}{\omega} d\omega = \pi$$

$$\text{or, } \int_0^{\infty} \frac{\sin \omega}{\omega} d\omega = \frac{\pi}{2}$$

$$\text{Therefore, } \int_0^{\infty} \frac{\sin t}{t} dt = \frac{\pi}{2}$$

(Ans.)

Relationship Between Fourier Series, Fourier Transformation & Fourier Analysis:

Fourier Series	Fourier Transformation	Fourier Analysis	
		Fourier Analysis	Fourier Synthesis
Deals with periodic functions/signals.	Deals with both periodic and aperiodic functions/signals.	Breaks signals into its constituent parts.	Synthesizes signals into its constituent parts.
Converts a periodic function into an infinite series of sine and/or cosine.	Converts time domain signal into frequency domain and vice-versa.		

Example 26: If $f(t) = \begin{cases} 1 - t^2 & \text{for } |t| < 1 \\ 0 & \text{for } |t| > 1 \end{cases}$; find the Fourier Transformation of $f(t)$.

Soln.: We know,

$$\begin{aligned}
 F.T. \{f(t)\} &= F(\omega) = \int_{-\infty}^{\infty} f(t) \cdot e^{i\omega t} dt \\
 &= \int_{-\infty}^{-1} f(t)e^{i\omega t} dt + \int_{-1}^1 f(t)e^{i\omega t} dt + \int_1^{\infty} f(t)e^{i\omega t} dt \\
 &= 0 + \int_{-1}^1 f(t)e^{i\omega t} dt + 0 \\
 &= \int_{-1}^1 (1 - t^2) \cdot e^{i\omega t} dt \dots \dots \dots (i)
 \end{aligned}$$

Now,

$$\begin{aligned}\int (1 - t^2) \cdot e^{i\omega t} dt &= (1 - t^2) \cdot \frac{e^{i\omega t}}{i\omega} - \int (-2t) \frac{e^{i\omega t}}{i\omega} dt \\&= (1 - t^2) \cdot \frac{e^{i\omega t}}{i\omega} + \frac{2}{i\omega} \left[t \cdot \frac{e^{i\omega t}}{i\omega} - \int \frac{e^{i\omega t}}{i\omega} dt \right] \\&= \frac{(1 - t^2)e^{i\omega t}}{i\omega} - \frac{2te^{i\omega t}}{\omega^2} + \frac{2e^{i\omega t}}{i\omega^3}\end{aligned}$$

From equation (i),

$$\begin{aligned}F.T. \{f(t)\} = F(\omega) &= \int_{-1}^1 (1 - t^2) \cdot e^{i\omega t} dt \\&= \left[\frac{(1 - t^2)e^{i\omega t}}{i\omega} - \frac{2te^{i\omega t}}{\omega^2} + \frac{2e^{i\omega t}}{i\omega^3} \right]_{-1}^1 \\&= 0 - \frac{2e^{i\omega}}{\omega^2} - \frac{2e^{-i\omega}}{\omega^2} + \frac{2e^{i\omega}}{i\omega^3} - \frac{2e^{-i\omega}}{i\omega^3} \\&= -\frac{2}{\omega^2} (e^{i\omega} - e^{-i\omega}) + \frac{2}{\omega^3} \cdot \frac{(e^{i\omega} - e^{-i\omega})}{i} \\&= -\frac{2}{\omega^2} \cdot 2 \cos \omega + \frac{2}{\omega^3} \cdot 2 \sin \omega \\&= \frac{4 \sin \omega}{\omega^3} - \frac{4 \cos \omega}{\omega^2} \\&= \frac{4}{\omega^2} \left(\frac{\sin \omega}{\omega} + \cos \omega \right) \\&\quad \text{(Ans.)}\end{aligned}$$

End of Part 2: Fourier Analysis